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# Source - https://stackoverflow.com/a
# Posted by tanemaki, modified by community. See post 'Timeline' for change history
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import pandas as pd
import numpy as np
from scipy import stats

df = pd.DataFrame(np.random.randn(100, 3))

df[(np.abs(stats.zscore(df)) < 3).all(axis=1)]#many standard deviations a data point is from the mean of its dataset.Z-score = (
```

	0	1	2
0	0.221433	-0.061200	-0.471501
1	1.342186	1.025072	-1.583382
2	1.147570	-0.167526	-2.088611
3	0.257703	-0.838250	1.335826
4	0.839495	-0.414140	0.475150
...	...	...	...
95	0.231537	1.693573	-0.027324
96	0.354629	-0.253889	1.125580
97	-1.003032	0.996137	1.001203
98	1.179589	-1.268854	0.677135
99	-0.669602	0.546432	1.437474

98 rows × 3 columns

```
import pandas as pd
import numpy as np
from scipy import stats

# Step 1: Create a simple dataframe with obvious outliers
data = {
    'Score': [85, 90, 78, 92, 88, 200, 85, 89, -50, 87]
}
df = pd.DataFrame(data)

print("ORIGINAL DATA:")
print(df)

# Step 2: Remove outliers using Z-score method
df_clean = df[(np.abs(stats.zscore(df)) < 3).all(axis=1)]

print("\nCLEANED DATA (outliers removed):")
print(df_clean)

# Step 3: Show what happened
print(f"\nSUMMARY:")
print(f"Original rows: {len(df)}")
print(f"Rows after cleaning: {len(df_clean)}")
print(f"Outliers removed: {len(df) - len(df_clean)}")
```

```
ORIGINAL DATA:
Score
0      85
1      90
2      78
3      92
4      88
5     200
6      85
7      89
8     -50
9      87
```

```
CLEANED DATA (outliers removed):
```

```
Score  
0      85  
1      90  
2      78  
3      92  
4      88  
6      85  
7      89  
9      87
```

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SUMMARY:
```

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Original rows: 10
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Rows after cleaning: 8
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Outliers removed: 2
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